

4.

Implementation

The five priority action areas of *ESD for 2030* are to be implemented by Member States and regional and global stakeholders with the support and assistance of UNESCO and its partners

4.1. Implementing *ESD for 2030* at country level



In implementing activities for the priority action areas, Member States are invited to mobilise the concerned stakeholders working in the five areas and support their collaborative networking under a coordinated strategy, related to the national framework on the SDGs. There shall also be more concrete efforts to develop communication and advocacy actions to reflect the SDGs in educational practices and frameworks. Nationwide multi-stakeholder initiatives can be set up to support *ESD for 2030* at the country level.

(Para 5.19, Framework for the implementation of *ESD for 2030*)

Objective

In order to create momentum to implement *ESD for 2030* and its five priority action areas, Member States are encouraged to engage in country initiatives to mainstream ESD in education and sustainable development.

Suggested country level activities

To implement *ESD for 2030* at country level, Member States are invited to:

- Engage in country initiatives on *ESD for 2030* that have or potentially could have country-wide impact. They can build on existing initiatives related to ESD or create a new initiative if necessary, in the context of SDG 4 contributions and expand them where possible. Typically, a country initiative would be initiated and led by the government, or by a non-governmental actor in collaboration with government.
- Integrate *ESD for 2030* activities into national frameworks on the SDGs, education or other relevant frameworks in order to ensure coherent action at country level.
- Address one or more of the five priority action areas through multiple activities that can be included as a part of one umbrella multi-stakeholder country initiative per country.
- Promote multi-sectoral partnerships and collaboration in particular with the key stakeholders of the five priority action areas in order to address all relevant education and sustainable development sectors. A working group at the country level should be responsible for overall coordination of planning, implementation, networking, monitoring, reporting, and communication with UNESCO. A national steering body on the SDGs, National Commissions for UNESCO, or relevant ministries and institutions may host the working group.
- Strengthen communication and advocacy efforts on the SDGs in order to mobilize populations beyond sustainable development communities.
- Monitor and report on the progress of country initiatives, with a clear set of targets, where possible.

UNESCO's activities

- UNESCO supports country initiatives on *ESD for 2030* based on the proposals submitted by Member States.
- UNESCO supports partnership and collaboration between country initiatives.
- UNESCO provides dedicated technical support to the identified country initiatives and organizes regional information meetings to provide technical guidance on their establishment.
- UNESCO monitors the progress of identified country initiatives to be included in global reporting of *ESD for 2030*. (See page 48 for more information).
- Country initiatives are expected to be self-funded. UNESCO may provide seed funding for country initiatives of developing countries.

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4.2. Harnessing partnership and collaboration



In implementing activities for the priority action areas, Member States are invited to mobilise the concerned stakeholders working in the five areas and support their collaborative networking under a coordinated strategy, related to the national framework on the SDGs.

UNESCO will continue supporting the networking of the key partners identified at the global level and their networking process. This will be done by providing them with regular platforms to meet and exchange information and experiences and plan initiatives collaboratively. The platforms will necessarily include representatives and actors from both government and private sectors and from the donor, development and SDG communities, as well as stakeholders working in the five priority action areas at national, regional and global levels.

(Para 5.19, 5.22, Framework for the implementation of *ESD for 2030*)

Objective

In order to implement the five priority action areas of *ESD for 2030*, networking opportunities among Member States and other stakeholders are organized to foster cross-national and cross-sectoral collaboration and create synergies.

Suggested country level activities

To harness partnership and collaboration, Member States are invited to:

- Strengthen partnership and collaboration among formal, non-formal and informal learning sectors, including intergenerational lifelong learning taking place in the community.
- Develop and support partnership and collaboration among the education, sustainable development and SDG including SDG 4 stakeholders.
- Reinforce public-private sector partnerships and collaboration to achieve structural changes required for a sustainable future and mobilize resources to support the process.
- Develop country level networks of *ESD for 2030* stakeholders to coordinate activities of country initiatives on *ESD for 2030*, and facilitate partnership and collaboration among the diverse stakeholders involved.
- Align and strengthen coordination and cooperation at global, regional, national, sub-national and local level concerning policies and practices on ESD to ensure mutually supportive synergies.

UNESCO's activities

ESD for 2030 Global Network (*ESD-Net*)

UNESCO establishes the *ESD for 2030* Global Network (*ESD-Net*):

- To facilitate dialogue, partnership and collaboration between representatives of country level initiatives and other partner organizations.
- To share lessons learned and best practices related to the implementation of *ESD for 2030* to provide peer-to-peer learning opportunities.
- To draw attention to ESD's on-going contribution to the SDGs as part of global advocacy on ESD.
- To report on and monitor progress towards the implementation of *ESD for 2030*.

Who are the members?

- Representatives of country level initiatives including governmental and non-governmental organizations; individual institutions; UN partners; members of development communities and donor agencies.

How does the network operate?

- UNESCO convenes a global meeting of the *ESD-Net* every two years, with the first foreseen in 2023, to discuss the latest progress on the implementation of *ESD for 2030*.
- In between meetings, an interactive learning environment will be facilitated online to create synergies among the members of the network.
- Regional, thematic or issue-based sub-groups of the network may be set up depending on identified needs. The work of regional sub-groups should be carried out in ways that maximize synergies with regional SDG 4 coordination mechanisms and international processes.

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4.3. Communicating for action



Meanwhile, with the presence of the SDGs announced specifically, there could also be a more proactive role for ESD to play for the achievement of the SDGs, with explicit reference to them. Communication and advocacy activities related to the SDGs in educational settings are good examples. They will typically involve informing learners of all ages of the existence of the 17 SDGs, and the implications of these goals for their individual and group lives, including the responsibilities individuals and institutions have to assume to help achieve them. These ESD activities are in great demand, as there is a need to reach out to learners of all ages and the general public to broaden their understanding of the SDGs and education's role in achieving them.'

'There will have to be more concentrated communication and advocacy efforts on the role of education in supporting the achievement of the SDGs. At the national level, this will be part of the nationwide initiatives for *ESD for 2030*. At the global level, UNESCO will develop a dedicated programme to inform concerned communities about the SDGs and learners and educators about the crucial role education can play in supporting the achievement of the SDGs. The UNESCO-Japan Prize programme, if it continues, will be an integral part of the UNESCO communication and advocacy strategy.

(Para 5.4, 5.24, Framework for the implementation of *ESD for 2030*)

Objective

In order to convey the urgency to act on the five priority action areas and ensure the success of *ESD for 2030*, Member States and other stakeholders are encouraged to conduct communication, outreach and advocacy activities, engaging a broader audience and a wider group of education and sustainable development partners.

Suggested country level activities

In relation to advocacy, Member States are invited to:

- Promote the role of education as a catalyst to meet all SDGs through targeted communication and outreach in all their ESD activities, mobilizing education stakeholders and the media.
- Emphasize the importance of education in their communication, outreach and advocacy work related to sustainable development and the SDGs, including public campaigns.
- Use innovative ways of communicating about their work on ESD, including digital fora, and prioritize engagement with young people.

UNESCO's activities

Empower key stakeholders in their communication efforts

UNESCO works with the key actors of the five priority action areas, namely education institutions, educators and practitioners, youth and policy-makers in order to enable them to inform and advocate for ESD among their stakeholders. UNESCO works to:

- Showcase best practices on ESD's contribution to the SDGs through its communication products and channels, including through the UNESCO Associated Schools Network
- Highlight outstanding ESD initiatives contributing to the achievement of the SDGs through the UNESCO-Japan Prize on ESD.
- Provide learning material and resources on the SDGs, featuring one or a selection of SDGs, highlighting relevant learning objectives and approaches, as well as inter-linkages and tensions between certain goals.
- Convene biennial global meetings of *ESD-Net* as a major platform for global advocacy for ESD as well as a network opportunity for relevant stakeholders.
- Support online channels to connect country initiatives and their stakeholders and organize joint days of action to create momentum and foster action on ESD at the global level.
- Build on the gains of UNESCO's 2020 campaign on ESD, engaging a global wider audience to encourage individuals and communities to take responsibility and action for sustainable development.
- Engage in UN and other international processes on sustainable development (such as Climate Change Conferences of the Parties) as opportunities to shine a spotlight on *ESD for 2030* and in collaboration with UN entities such as UN DESA, UNEP One Planet Network and others responsible for the management of the SDGs, as well as other global partners such as Regional Centres of Expertise on ESD.

Advocate at the global level

UNESCO also works to highlight the role of ESD at the global level, in particular its role as a key enabler for the achievement of all 17 SDGs. UNESCO works to:

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4.4. Tracking issues and trends



The implementation of *ESD for 2030* should be evidence-informed. Emerging issues and trends need to be monitored closely and analysed for their pedagogical implications. To that effect, UNESCO is suggested to conduct analytical work on issues identified, especially those related to individuals' transformational process, structural issues underlying development and sustainability and the future of ESD in the era of technological advances, with the results disseminated to the actors on the ground.

(Para 5.23, Framework for the implementation of *ESD for 2030*)

Objective

In order to support the implementation of the five priority action areas of *ESD for 2030* with concrete evidence and strategic foresight, the latest developments in education and sustainable development sectors are to be reviewed.

Suggested country level activities

To track issues and trends, Member States are invited to:

- Encourage a review of the 17 SDGs within local, national, and regional contexts in education settings as part of ESD activities to identify relevant issues, trends and questions that need to be further examined and discussed.
- Conduct thematic research related to ESD and the SDGs, based on the key issues, trends and questions raised in relation to the five priority action areas. Higher education and research institutions should play a key role in this type of research activity, which can be included as a part of country initiatives on *ESD for 2030*.
- Disseminate and review the outcomes of research to create learning opportunities to raise critical questions, in particular on the inter-linkages and tensions between the different SDGs.
- Collaborate with other countries to review sustainable development and ESD-related emerging issues and trends in specific contexts, in order to find appropriate educational responses.

UNESCO's activities

UNESCO conducts thematic research, expert consultations, and publishes relevant reports on emerging issues and trends on key themes related to ESD's contribution to the SDGs. Together with its partners, UNESCO:

- Conducts expert consultations to identify and review key emerging issues and trends;
- Carries out studies to collect and analyse relevant evidence from Member States;
- Produces evidence-based reports in order to mobilize the actors of *ESD for 2030*.

- Disseminates the latest knowledge products through advocacy channels including biennial global meetings and training programmes.

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THE THEMATIC FOCUS AREAS

Special attention will be paid to the themes and potential questions related to key contexts and reflections of *ESD for 2030*, including but not limited to:

ESD and the SDGs

- What are successful pedagogical practices to achieve the 17 SDGs?
- How can ESD address the inter-linkages and tensions between and among different SDGs?

Transformative action and ESD

- How does individual transformation take place and how can education facilitate a tipping point for a learner to undertake transformative actions for sustainability?
- What is the role of communities to promote transformative action for sustainable future?

Structural issues and ESD

- How can ESD support the balancing act between economic and sustainable development?
- How can ESD encourage learners to explore alternative values to consumer societies?
- In the context of extreme poverty, what kind of ESD best connects with and supports learners' daily challenges?

Technological advances and ESD

- What are the implications of artificial intelligence and the 4th Industrial Revolution for ESD?
- How can ESD address the risks and benefits of green technologies?
- How can new technologies be harnessed to educate for sustainability?

4.5. Mobilizing resources



Also required are partnerships with other key players, such as multilateral financial institutions, national development agencies, the private sector, civil society and community actors. The alliance with national mechanisms set up to support the achievement of the SDGs through education will also need further attention. In developing partnerships, the use of innovative and creative financing mechanisms will deserve particular attention.'

'Within UNESCO, the Education Sector shall further strengthen its inter-sectoral partnership with other Sectors, especially Culture and Science, integrating the implementation of *ESD for 2030*, where possible, into their relevant programmes, which include, among others, World Heritage sites, intangible cultural heritage, the network of Biosphere Reserves, Geoparks, Small Island Developing States, and local and indigenous knowledge. The Education Sector should also further enhance its intra-sectoral collaboration on ESD and encourage the contribution of relevant programmes and networks to *ESD for 2030*, including the UNESCO Associated Schools Network (ASPnet), UNESCO Chairs, UNEVOC Centres on Technical and Vocational Education and Training (TVET), the Learning Cities Network, as well as the Education 2030 Steering Committee.

(Para 5.25-5.27, Framework for the implementation of *ESD for 2030*)

Objective

In order to implement effectively the five priority action areas of *ESD for 2030*, mobilization of necessary resources is crucial.

Suggested country level activities

To mobilize resources for *ESD for 2030*, Member States are invited to:

- Explore existing funds associated with education to finance activities of *ESD for 2030* such as the budget and programme on school support, teacher training, youth empowerment, lifelong learning in cities and communities, with the understanding that ESD is not a topic but a holistic approach to reshaping education to address sustainability challenges.
- Tap into existing resources allocated on sustainable development, national implementation of the SDGs, as well as international cooperation to create synergies between ESD and relevant areas.
- Consider dedicating new budgets for *ESD for 2030* as a part of a national drive for quality education and its contribution to the collective survival and prosperity of humanity.
- Utilize relevant regional initiatives and their resources to make collaborative efforts with other countries on *ESD for 2030*.
- Mobilize resources in the private sector, in particular to transform TVET institutions and private company training centres to support skills development in green-related field areas and become active leaders of *ESD for 2030*.
- Find creative solutions to implement effectively *ESD for 2030* without major additional resources, such as embedding ESD into education quality assurance criteria to assess the performance of education institutions.

UNESCO's activities

- UNESCO's mobilization of resources for Member States' implementation of *ESD for 2030*
- At global level, UNESCO will make efforts to identify available funds for education and sustainable development, and advocate for use of these funds to support activities of *ESD for 2030*.
- All activities of *ESD for 2030*, including country initiatives are expected to be self-funded. UNESCO will provide some seed funding to a limited number of initiatives from developing countries, in particular the least developed countries.
- Resources for UNESCO's activities on *ESD for 2030*
- UNESCO will continue to make efforts to mobilize voluntary contributions.
- UNESCO will leverage in-kind support from UNESCO networks for the implementation of *ESD for 2030*, in particular, ASPnet, UNESCO Chairs, UNESCO centres including UNEVOC Centres on TVET, the World Network of Biosphere Reserves and Geoparks, World Heritage sites and intangible cultural heritage, and UNESCO Cities networks (notably the Global Learning Cities Network and the UNESCO Creative Cities Network).

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4.6. Monitoring progress



Various types of monitoring and evaluation will remain as one of the important tasks of UNESCO as outlined below. In light of the lessons from the GAP, concrete monitoring measures must be pursued and improved with the aim of achieving SDG target 4.7 by 2030.

First, the scaling-up of actions in the five priority action areas by the Member States as well as by the identified Partners at global level needs close monitoring. [...]

For the measuring of scaling-up in the Member States, UNESCO will revise the existing survey questionnaire for the monitoring of Target 4.7. It is proposed to integrate the effort to measure progress of countries' scaling-up of actions in the five priority action areas into this instrument. [...]

The second type of monitoring will be carried out within the defined scope of a programme or a project, the focus of which will be to ensure the production of the planned outputs and the spread of their impact. [...]

Furthermore, there will be a new monitoring effort made by UNESCO. In order to support the evidence-informed principle underpinning the implementation of the post-GAP position, UNESCO will carry out regular thematic surveys to identify and analyse key issues, trends and developments. [...]

Throughout the implementation of *ESD for 2030*, periodic evaluations of the progress of the implementation of the framework will be undertaken

(Para 5.28-5.33, Framework for the implementation of *ESD for 2030*)

Objective

In order to strategically and effectively steer the actions undertaken on the five priority action areas, systematic assessment of the progress of *ESD for 2030* will take place.

Suggested country level activities

To monitor the progress of *ESD for 2030*, Member States are invited to:

- Develop national indicators to monitor the progress of dedicated country initiatives on *ESD for 2030* and contribute to the development of a meaningful scheme to monitor the outcome of ESD.
- Regularly monitor and evaluate the progress on ESD at country level, in particular by conducting surveys to assess the progress of the five priority action areas.
- Report on ESD as part of global and regional SDG monitoring including that of SDG Target 4.7.

UNESCO's activities

UNESCO carries out periodic evaluation of the progress of the implementation of *ESD for 2030* on the basis of data and information collected through the following four activities:

- monitoring the overall progress of ESD Member States;
- monitoring the progress in the five priority action areas;
- monitoring the implementation of planned activities through country initiatives and others;
- reviewing additional data on the progress of ESD.
- UNESCO plans to conduct two evaluations in 2020-2030: A mid-term evaluation in 2025 and a final evaluation in 2030. Additional shorter reports are to be published in between.

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KEY INDICATORS TO MONITOR THE PROGRESS OF *ESD FOR 2030*

Policy

Extent to which legal frameworks and policies are in place to promote ESD.

Learning environment

Extent to which learning environment promotes ESD and its whole-institution approach.

Educators

Extent to which educators are trained to be able to teach ESD and apply whole-institution approaches to ESD in learning situations.

Youth

Extent to which youth are engaged in ESD.

Community

Extent to which ESD is promoted in local communities.

Progress of country initiatives:

Extent to which *ESD for 2030* is implemented in countries around the world.

Table 1. Monitoring and Evaluation Framework of *ESD for 2030*

	What	Who
Overall progress in ESD	UNESCO as the custodian agency for the global indicator 4.7.1 of SDG Target 4.7, will use data collected through the 1974 Recommendation reporting process to monitor Member States' progress on ESD.	UNESCO with Member States
Progress of <i>ESD for 2030</i> implementation through country initiatives	The achievements in the five priority action areas by country initiatives on <i>ESD for 2030</i> will be monitored.	UNESCO with Member States
Progress of <i>ESD for 2030</i> implementation through the activities of other partner organizations of ESD-Net	The achievements in the five priority action areas by the <i>ESD for 2030</i> activities of other partner organizations of <i>ESD-Net</i> will be monitored.	UNESCO with other partner organizations of <i>ESD-Net</i>
Additional quantitative and qualitative information on ESD progress and impact	In addition to formal monitoring, UNESCO will conduct selective monitoring efforts to capture evidence on the impact of <i>ESD for 2030</i> .	UNESCO

How

Indicators

Every four years, Member States report to UNESCO on the implementation of 'the Recommendation concerning Education for International Understanding, Cooperation and Peace and Education relating to Human Rights and Fundamental Freedoms' adopted at the 18th UNESCO General Conference in 1974. The findings of the progress on the 1974 Recommendation provide important insights into the progress towards Target 4.7 of SDG 4 on Education, which concerns ESD among others.

Global Indicator 4.7.1: Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment

UNESCO will support Member States involved in *ESD for 2030* to set their targets and baselines in each of the five priority action areas to which they plan to contribute.

Countries will be invited to participate in regular surveys every two years to measure relative progress on the five priority action areas against the baseline set at the outset. The result of the survey will be discussed at the biennial Global Partners Meetings.

Following each biennial review, stakeholders will be invited to reassess and progressively increase their targets in the course of 2020-2030.

The biennial surveys will monitor the progress of one indicator per priority action area and one indicator on the general progress of *ESD for 2030*. (See the list on page 49)

UNESCO will support other partner organizations in *ESD-Net* to set their targets and baselines in each of the five priority action areas to which they plan to contribute.

Partner organizations will be invited to participate in regular surveys every two years to measure relative progress on the five priority action areas against the baseline set at the outset. The result of the survey will be discussed at the biennial Global Partners Meetings.

Following each biennial review, stakeholders will be invited to reassess and progressively increase their targets in the course of 2020-2030.

The biennial surveys will monitor the progress of one indicator per priority action area and one indicator on the general progress of *ESD for 2030*. (See the list on page 49)

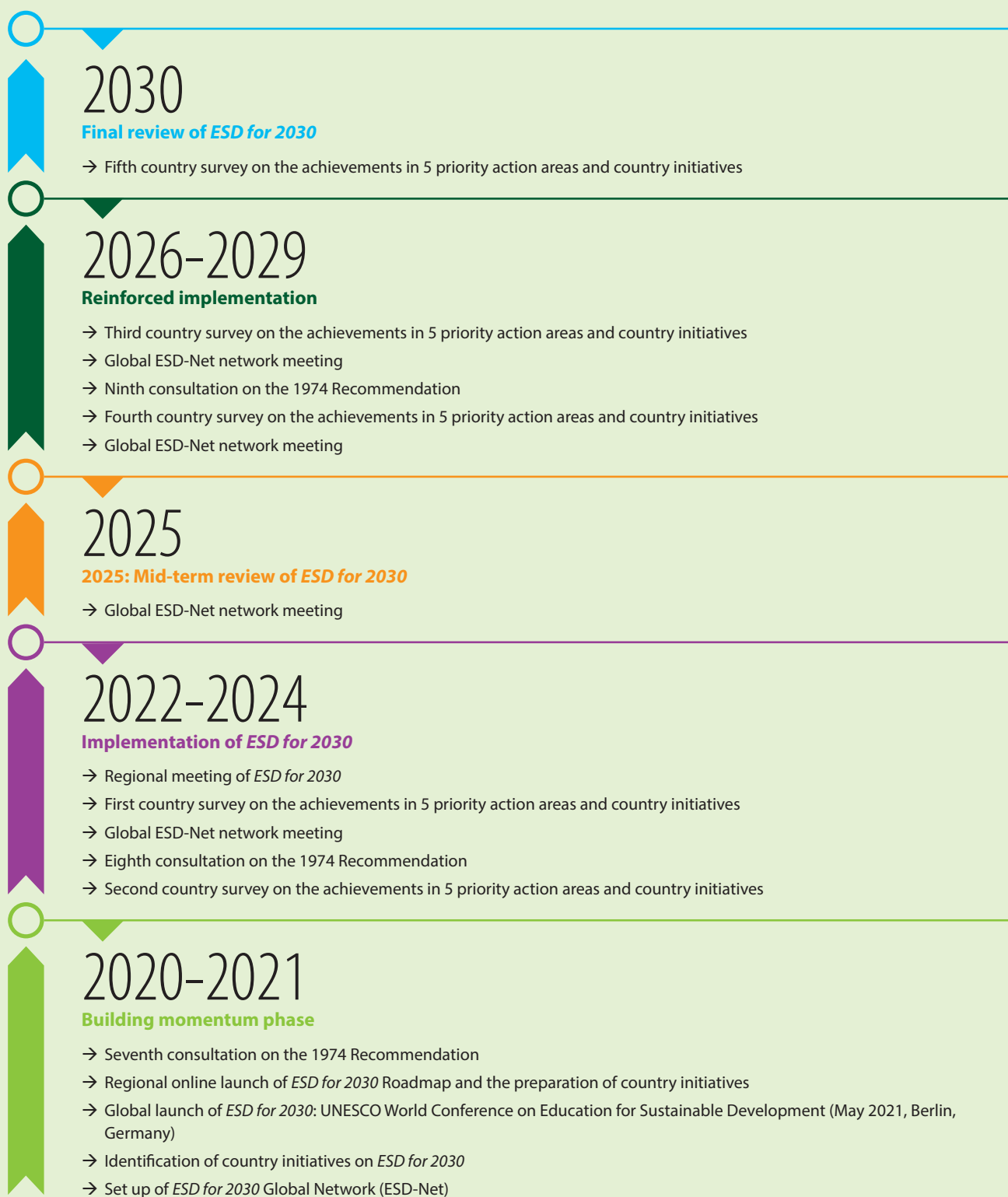
UNESCO will conduct secondary analyses of existing evidence and data, in collaboration with other relevant UN agencies and partners to gain further insights on the progress of ESD.

Thematic surveys will be conducted to identify and analyse key issues, trends and developments, to complement the normative monitoring of Target 4.7. (See section 4 on tracking issues and trends)

UNESCO will continue to monitor the qualitative impact of ESD by collecting and highlighting testimonies and best practices.

N/A

4.7. Planning for the next 10 years



Annexes

A. One-page summary of *ESD for 2030* framework



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B. Framework for the implementation of Education for Sustainable Development (ESD) beyond 2019: *ESD for 2030*

Education for Sustainable Development: Towards achieving the SDGs (*ESD for 2030*)

Framework for the implementation of Education for Sustainable Development beyond 2019

1. Why this framework?

- 1.1 In 1992, the United Nations Conference on Environment and Development held in Rio de Janeiro recognized, in its framework for action Agenda 21, the critical role that education can play in the transition to sustainable development. In 2002, the United Nations General Assembly declared the ten years from 2005 to 2014 to be the Decade of Education for Sustainable Development (DESD), calling on governments to integrate the principles of sustainability into their educational strategies and action plans. UNESCO, which had been assigned the role of task manager for Chapter 36 on education of Agenda 21, became the UN-designated lead agency.
- 1.2 In 2014, the DESD came to an end and, as the follow up, UNESCO launched⁸ the Global Action Programme (GAP) on Education for Sustainable Development for an initial phase of five years (2015-2019) at the UNESCO World Conference on Education for Sustainable Development held in Aichi-Nagoya, Japan. At the same time, the Aichi-Nagoya Declaration on Education for Sustainable Development was adopted recognizing ESD's role in empowering learners and the societies they live in.⁹ To build on the efforts made by the DESD for advocacy and awareness-raising,

the GAP, acknowledged by the United Nations General Assembly¹⁰ as the official follow-up to the DESD, focused on the scaling up of actions on the ground. Advancing policy, transforming learning and training environments, building capacities of educators and trainers, mobilizing youth, and accelerating sustainable solutions at local level were identified as the five priority action areas for implementation.

- 1.3 The five-year cycle of the GAP is due to end in 2019, and in order to build a post-GAP position that contributes to Agenda 2030 thereafter, this framework was being prepared.

2. How this framework has been prepared

- 2.1 **Review of the GAP:** The review of GAP implementation, covering the period 2015 to 2018¹¹, revealed that good progress was being made in the implementation of the targeted objectives set by the Key GAP Partners.¹² In the additional consultation held with Key GAP Partners and others, however, the need for more inter-linkages between partners working for different priority action areas was voiced. In the internal review of the mid-term implementation, the insufficient visibility of the engagement made by governments was also pointed out.
- 2.2 **Symposium series:** In order to identify issues and trends that need to be considered and updated in the post-GAP framework, UNESCO planned and organized four brainstorming symposia with more than 250 experts and stakeholders involved. They were held in Brazil, Germany, Japan and South Africa from 2016 to 2018, and comprised of interviews and visits to concrete actors and their practices on the ground. The discussions focused on what made people living under different contexts act in favour of sustainable development and provided

⁸ As endorsed by the 37th session of the UNESCO General Conference ([87 C/Resolution 13](#)).

⁹ The Aichi-Nagoya Declaration on Education for Sustainable Development states ESD can empower learners to transform themselves and the society they live in by developing knowledge, skills, attitudes, competencies and values required for addressing global citizenship and local contextual challenges of the present and the future, such as critical and systemic thinking, analytical problem-solving, creativity, working collaboratively and making decisions in the face of uncertainty, and understanding the interconnectedness of global challenges and responsibilities emanating from such awareness.

¹⁰ Resolution A/RES/69/211.

¹¹ The final evaluation of the GAP implementation will be carried out by the end of 2019. If the result of the final evaluation was to be fully integrated into the post-GAP framework, there would be a gap of about one year before the new post-GAP framework is launched, which requires institutional processes involving the Governing Bodies of UNESCO and the United Nations. In order to avoid the break in implementation, and also as the results of the mid-term review showed a progress trend not expected to be reversed in any way, the process to prepare this post-GAP position has been advanced. This advanced process is supported by [E04 EX/Decision 28](#) of UNESCO's Executive Board, which calls for 'seamless follow-up to the Global Action Programme and continued UNESCO leadership in ESD beyond 2019'.

¹² 2018 GAP survey: <https://en.unesco.org/themes/education-sustainable-development> According to the GAP Key Partner Survey (2015-2018), it is estimated that GAP partners (and their cascaded networks of partners) supported over 900 strategic policy developments, while more than 1,400 programmes were developed and implemented to support countries' policy development on ESD. Around 151,000 schools and 26 million learners were estimated to have been supported by the partners along with 2 million educators and 48,400 teacher training institutions. The partners also supported around 3.4 million youth leaders and trained around 762,000 as trainers. Finally, it is projected that over 5,600 civil society organizations conducted ESD activities and that over 2,300 local ESD activities were established.

important insights into the observations described in the Required Reflections part of this framework.

- 2.3 **Consultation process:** Based on the observations drawn from the above processes and several internal reviews of UNESCO's ongoing ESD activities, UNESCO prepared a zero draft of this framework. The first version was shared with and reviewed by GAP Key Partners during a meeting organized by UNESCO in April 2018 in San Jose, Costa Rica. The second version was submitted to a Technical Consultation Meeting with government representatives and experts from 116 Member States, held in July 2018 in Bangkok, Thailand. A wider online consultation with United Nations agencies and other international bodies as well as an online public consultation with a wide range of stakeholders were carried out up until the end of 2018.

3. Key contexts

- 3.1 **DESD and the GAP:** The DESD aimed at integrating the principles and practices of sustainable development into all aspects of education and learning. Its main contribution was awareness-raising, which helped generate interesting practices and projects. The evaluation of the DESD recommended that, among others, institutional support is needed in order to scale up and maintain these momentous initiatives.
- 3.2 The GAP (2015-2019) filled these shortcomings. Actors working on policy, education and training institutions, educators, youth and local authorities were identified as Key Partners to work in one of the five priority action areas – advancing policy, transforming learning and training environments, building capacities of educators and trainers, empowering and mobilizing youth, and accelerating sustainable solutions at local level. Their sectoral networking was supported by the five Partner Networks set up for the five areas.
- 3.3 The mid-term review of GAP implementation in 2017 reported on the good progress that the Key Partners were making in meeting their scaling-up targets. The networking mechanism was found to be useful for mutual learning, but less so for raising funds and developing

joint projects. Lessons learned also include that networking efforts must be combined with concrete programme activities that bring Partners together, and that Partners of different sectors should work more collaboratively across the priority action areas.

- 3.4 Furthermore, although policy development has been one of the priority action areas and policy-makers have been active in the implementation of GAP through their participation in the Partner Network for policy development, the leadership assumed by governments was not visibly demonstrated in the current phase of GAP implementation. This was partly due to the fact that global monitoring has focused mainly on the totality of actions scaled up on the ground, which are not necessarily categorized by the type of actors.
- 3.5 This is regrettable as data actually show that governments are indeed making increasing efforts to reflect sustainability principles in the policy arena. According to a survey UNESCO conducted in 2016¹³, for example, over 91% of the responding Member States reported on the increase of their emphasis on ESD in the policy arena. In the post-GAP framework, therefore, there has to be a special window for monitoring and reporting on the leadership taken by government actors.
- 3.6 **SDGs:** In 2015, the global community launched 17 Sustainable Development Goals (SDGs) addressing issues related to poverty, hunger, health, education, energy, work, industry, inequalities, cities, consumption, climate, ocean life, ecosystems, peace and partnership. Achieving these goals requires a profound transformation in the way we live, think and act; and the role of education in achieving all of the 17 SDGs is given heightened emphasis.
- 3.7 ESD has advocated for the sustainability principle to be reflected in education policy, curricula, training and practice as a means to empower individuals to take informed decisions. However, some further critical reflection is needed around the fact that, despite its advocacy on a holistic and system-wide approach, when it comes to implementation, ESD tends to be treated as a thematic topic. This has to change and ESD has

13 As part of the 6th Consultation on the 1974 UNESCO Recommendation concerning Education for International Understanding, Cooperation and Peace and Education relating to Human Rights and Fundamental Freedoms, which includes questions on ESD. The survey tool used for the monitoring of the 1974 Recommendation is also the one used for monitoring Target 4.7 of SDG4, where ESD is an integral element.

to outgrow its topical understanding and work more proactively at the systemic level as a part of SDG 4 on education, and Target 4.7 in particular.

- 3.8 ESD and Global Citizenship Education (GCED) are at the core of SDG 4 Target 4.7, which aims to ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development.
- 3.9 The SDGs provide an excellent opportunity to strengthen our perspective on ESD. UN General Assembly Resolution 72/222¹⁴ recognized ESD as 'an integral element of SDG 4 on education and a key enabler of all the other SDGs'. The 17 SDGs cover the key environmental challenges (e.g. climate change, water, ocean and land), and also the fundamental issues that need to be taken into consideration in development discourses, such as poverty, gender equality¹⁵ and equity.¹⁶
- 3.10 The criticism that ESD has received for its 'vagueness', which stems in part from its broad and inclusive perspective and the ever-changing nature of sustainability issues, can be addressed more concretely when the field is linked with the SDGs. Among the 17 SDGs, prioritization depending on national and local needs can be considered. For example, climate change remains one of the key focus areas. ESD can also address the interlinkages of the SDGs, long advocated for as essential for genuine progress towards sustainable development. The marriage of ESD and the SDGs is not only a necessity but a 'win-win' for both fields.

4. Required reflections

- 4.1 **Transformative action:** How to encourage learners to undertake transformative actions for sustainability has been a major preoccupation for ESD. The symposium series, held from 2016 to 2018 around the world in preparation of this document, revealed a few important insights into how transformative actions take place.
- 4.2 First of all, transformation necessitates, among other things, a certain level of disruption, with people opting to step outside the safety of the status quo or the "usual" way of thinking, behaving or living. It requires courage, persistence and determination, which can be present at different degrees, and which are best sourced from personal conviction, insight, or the simple feeling of what is right.
- 4.3 Second, there are different stages of transformation. With the acquisition of knowledge and information, learners come to be aware of the existence of certain realities. With critical analysis, they begin to understand the complexity of those realities. An experiential exposure to the realities provides them with a deeper connection with the issues, which can also lead to an empathic connection to those affected by the said realities. Empathy can turn into compassion if the exposed realities bear relevance to the learners' own lives and their sense of identity. A tipping point arrives where a compassionate mind is set on the path of empowerment.¹⁷
- 4.4 The pedagogical implications of this understanding are many and various. Opportunities to launch critical inquiry, exposure to realities, relevance to our own lives and the presence of influential peers, mentors or role models, as well as tipping points, play an important role in empowering individuals to take decisive actions. It is also important for

¹⁴ ESD is also part of various other international agreements concerned with sustainable development, such as the three 'Rio Conventions' (on climate change, biodiversity and desertification), the Paris Agreement on Climate Change, the Sendai Framework for Disaster Risk Reduction, and the 10-Year Framework of Programmes on Sustainable Consumption and Production (2012-2021), among others.

¹⁵ For example, ESD is an instrument to achieve all the SDGs, and each of the SDGs comes with specific gendered challenges. ESD takes on a cross-disciplinary and systemic approach that enables the question of gender equality to be linked to the various issues of sustainable development. There is, for example, a gendered facet of vulnerability to hazards induced by climate change. When disasters occur, more women die than men because social rules of conduct mean that, for example in the case of flooding, women often have not learned to swim, and have behavioural restrictions that limit their mobility in the face of risk. It should therefore become a priority to provide women with access to ESD. In this regard, ESD actively promotes gender equality, and creates conditions and strategies that empower women.

¹⁶ Note that the concept of sustainable development and its challenges include a vast array of complex issues and views that may not have been fully represented in the SDGs.

¹⁷ One does not necessarily go through the stages of transformation (awareness, understanding of complexities, empathy, compassion, empowerment) in a linear fashion. Stages can be skipped with individuals moving from, say, the first factual stage, directly to the final stage of empowerment. There can also be cases where individuals start with empathy or compassion, and only later on approach the matter with cognitive awareness, instead of starting with the awareness stage. The process and the pace at which individuals go through these stages can also vary.

learners to have the space to experiment with new 'disruptive' ideas, which can facilitate critical perspectives as well as bring about a tipping point. Formal education alone is not enough. Non-formal education and informal learning, including intergenerational lifelong learning taking place in the community, provide learners with critical opportunities to relate them to the realities that concern them and to be influenced to take the necessary actions.

- 4.5 There has to be more attention to individuals and how they are transformed. Fundamental changes required for a sustainable future start with individuals and their change of behaviour, attitude and lifestyle, while the contextual factors and institutional support provide an enabling environment and can bulwark individual contributions. This is particularly so among the younger generation whose transformative action is often prompted when they attach importance to certain values and a lifestyle that corresponds to their sense of identity. ESD is needed to provide them with critical thinking skills to reflect on individual values, attitudes and behaviours as well as lifestyle choices.
- 4.6 Last, but not least, the reflection on transformative action points to the absolute importance of community. In a community, which can be defined not only physically, but also virtually, socially, politically or culturally, learners find values and causes that concern them both individually and collectively. They can also find other fellow members and bond with them, which generates solidarity and facilitates collective action for transformation and a culture of sustainability.
- 4.7 ESD in action is basically citizenship in action. It evokes the lifelong learning perspective, taking place not only at school, but also outside the school environment, throughout the life of each individual. Based on human rights and principles such as participation, non-discrimination and accountability, it interacts with the social and cultural milieu of the community and stimulates social learning within it. Cultural identity can play an important role. In order to link up with communities more closely, schools should be granted more autonomy in implementing the

curricula framework and managing their daily activities. ESD in action therefore requires a new perspective on the roles and functions of schools.

- 4.8 **Structural changes:** As much as attention is required for what is happening at the individual level in relation to transformative decisions and experience, there is a need for ESD to focus more on deep structural causes.
- 4.9 The relationship between economic growth and sustainable development is one of the pertinent issues in this regard. There is wide agreement that it is challenging to reconcile economic growth with the principles of sustainable development, as far as current industrial and production patterns continue. Ever accelerating production and consumption deplete natural resources, produce unmanageable amounts of waste and lead to a rise in global temperatures. Many laudable and credible initiatives to promote sustainable production and consumption exist, but their impact has been limited.
- 4.10 ESD in the future will have to encourage learners to explore values which could provide an alternative to consumer societies, such as sufficiency, fairness and solidarity. The emerging interest in a circular economy and a sharing economy represents one of those alternatives.¹⁸ ESD can provide training for key stakeholders in the private sector and raise financial literacy to support sustainable economic practices. ESD also has to affect the unsustainable production patterns of current economic structures more directly. This means that people have to be empowered to engage directly in the political process and advocate, for example, for appropriate environmental regulations for businesses.
- 4.11 A structural view is required also to address ESD in the context of extreme poverty. Some consider that ESD is a 'luxury' in the context of extreme poverty or other challenging survival situations (e.g. conflict or refugee situations). There is some truth to this in that the full complexity and relevance of the concept of sustainable development does not immediately resonate with people trying to survive on a daily basis, but

¹⁸ A circular economy promotes a shift from a 'take, make and dispose' model to a 'cradle to cradle' approach based on long-lasting product design, repair, reuse, refurbishing and recycling. It argues that, even in a sustainable world, consumers can keep enjoying similar products and services. For example, mobile phones can be designed in a different way to allow valuable parts to be reused more easily. While massive amounts of clothes are thrown away due to the 'fast-fashion' trend, some textile companies are restructuring their operation systems to enable use of more recycled fibres in place of primary raw materials.

this does not negate the importance of ESD to populations in need.

- 4.12 Populations in extreme poverty are often the victims of calamitous development and natural disasters. They are affected much more directly by environmental degradation and the lack of economic and social sustainability. ESD approaches that may work for populations living in more fortunate situations may not necessarily be effective for populations in need. ESD approaches need to be contextualized to the realities of target populations.
- 4.13 For example, the use of community, advocated above as the platform for action, may be approached differently in contexts of extreme poverty. Extreme poverty is often rampant among migrant populations. The 'members' of these groups are brought together artificially with no inherent ties or shared identity. Populations in refugee or other conflictual situations often face the same fate. For such populations, community, which could provide a useful platform for action, should take into consideration the particular issue of group identity.
- 4.14 Furthermore, the approach for populations in extreme poverty may require more attention to the fundamentals. The starting point should be, first and foremost, to ensure and restore human dignity and the right to live decently. Providing people with basic and other relevant life-skills, or skills to ensure their livelihood in order to confront and overcome extreme poverty, is a priority. This cannot be done without addressing the wider political, historical, social and economic contexts from which their suffering springs and is perpetuated.
- 4.15 **The technological future:** According to a report published in 2015 by the World Economic Forum, by 2025, or within less than just ten years, the world will witness major technological shifts that will change every aspect of our societies. The tipping points will eventually change not only our lives and environments, but also our discourse on sustainability. Some 'old' problems will be resolved, but new challenges and risks will arise. ESD for the future cannot afford not to address the implications of the technological era.
- 4.16 The ubiquitous connection of the internet to our environments will help us better monitor and manage energy use. For example, smart city technologies will increase efficiency in managing energy, traffic and logistics. Innovations like 3-D printing may reduce the transportation required for the delivery of materials. Some of the long-standing ESD efforts to change people's behaviour in relation to energy-saving, resource management and a green environment may no longer be relevant. At the same time, new opportunities will open up for ESD. For example, ESD can accelerate the transition to green technologies through equipping people with the required green skills.
- 4.17 On the other hand, there will also be a surge of new challenges. For example, 3-D printing can result in the growth of waste for disposal and higher consumption. A digitally automated and connected world helps secure comfort and quality of life, but it can also pose its own security and safety risks and lead to new sorts of disasters, the impact of which could be even more widespread than in a less connected world. Applications of Artificial Intelligence are already strongly influencing human behaviour in consumption or social interaction.¹⁹ The unequal sharing of the benefits of technological progress can also enlarge the gap between the 'haves' and the 'have-nots'. ESD for the future should prepare the next generations to stay vigilant for 'new' problems.
- 4.18 Importantly, a critical perspective is required on the continuing relevance of 'traditional' sustainability values. With sensor-equipped buildings, the behaviour of switching off lights itself may become unnecessary and extinct, but the value of saving energy will and should remain relevant. Ironically, the task of teaching sustainability principles may become more challenging, as technologies give the illusion that they have resolved or can resolve the majority of sustainability problems.
- 4.19 Considering the fact that the business, manufacturing and enterprise sector as well as the scientific community are often the foremost actors and beneficiaries of technological innovations, it is of utmost importance for ESD to link up more closely with the stakeholders in these communities. Orienting ESD to support

19 The relationship between frontier technology, such as digitalization and Artificial Intelligence, and ESD should be explored.

the achievement of the SDGs will provide the opportunity for the ESD community to work more closely with the actors of the above communities, who are the main stakeholders of these various goals.

5. Implementation framework

- 5.1 **Title:** The proposed name for the framework to support the post-GAP position is Education for Sustainable Development: Towards achieving the SDGs (*ESD for 2030*). It aims to make clear the aspect of continuity from the current phase of the GAP to the post-GAP position, while highlighting its alignment with the 2030 Sustainable Development Agenda.
- 5.2 **Goal:** *ESD for 2030* aims to build a more just and sustainable world through strengthening ESD and contributing to the achievement of the 17 SDGs. *ESD for 2030* will achieve this through a threefold approach.
- 5.3 All ESD activities contribute to the achievement of the SDGs. ESD activities carried out by Member States and UNESCO, even without explicit reference to the SDGs, addressed many issues now concretized into specific SDGs. They include, among others, climate change, enshrined in SDG 13 on climate action, ocean life in SDG 14, biodiversity in SDG 15 as well as clean energy in SDG 7, sustainable cities and communities in SDG 11, responsible consumption and production in SDG 12, and education in SDG 4. The contributions of these long-standing ESD activities to the achievement of the SDGs are evident and they deserve continuing support.
- 5.4 Meanwhile, there could also be a more proactive role for ESD to play for the achievement of the SDGs, with explicit reference to them. Communication and advocacy activities related to the SDGs in educational settings are good examples. They will typically involve informing learners of all ages of the existence of the 17 SDGs, and the implications of these goals for their individual and group lives, including the responsibilities individuals and institutions have to assume to help achieve them. These ESD activities are in great demand, as there is a need to reach out to learners of all ages and the general public to broaden their understanding of the SDGs and education's role in achieving them.
- 5.5 ESD's linkages to the SDGs can go even deeper and beyond communication and advocacy. ESD activities might raise critical questions on the inter-linkages between and among different SDGs, which can involve tensions and lack of clarity. These types of ESD activities will not stop at just addressing topics related to the SDGs or with serving as the communication tool for the SDGs. They will find their *raison d'être* in raising critical questions related to development or sustainable development itself, with the inter-linkages between the different SDGs.
- 5.6 SDG 12 on responsible consumption and production, for example, needs to be addressed in connection with SDG 8 on decent work and economic growth and there are structural quandaries to be resolved between the two. SDG 9 on industry, innovation and infrastructure needs to consider issues related to SDG 11 on sustainable cities and communities and the achievement of both of these demands a balancing act. ESD in the *ESD for 2030* position should present a balancing perspective on development.
- 5.7 The attempt to raise critical and structural questions and view development as a balancing act is not new to the ESD community. From the DESD and throughout the implementation of the current phase of the GAP, reorienting societies towards sustainable development has remained as the ultimate aim of ESD. The 17 SDGs, which encompass the issues related to development and sustainability in a single framework, offer to the global community a renewed window of opportunity to reinforce this fundamental function of ESD.
- 5.8 While ESD contributes to all 17 SDGs, it continues to have particular relevance for the global education agenda enshrined in SDG 4, in which ESD has a distinct place. ESD is a key element of quality education. Its cross-cutting competencies in cognitive, socio-emotional and behavioural dimensions of learning bear relevance to all areas of education. Its particular emphasis on competencies related to empathy, solidarity and action-taking can help advance SDG 4 towards the future where education contributes not only to the successes of individuals, but also to the collective survival and prosperity of the global community. It will also help the global education agenda move away from an exclusive focus on access and quality measured mainly in terms

of learning outcomes, towards an increased emphasis on learning contents and their contribution to humanity.

- 5.9 **Structures:** The principal aim of the current phase of the GAP is to support the scaling up of actions on the ground. This objective, according to the mid-term review, is being achieved successfully by the GAP Key Partners, generating a large volume of activities on the ground. With the overall approach adopted by the current phase of the GAP proven effective, *ESD for 2030* recommends its main structural components be maintained, while a few adjustments and updates are suggested, based on lessons learned from the implementation.
- 5.10 First, the five priority action areas – namely, advancing policy, transforming learning and training environments, building capacities of educators and trainers, empowering and mobilising youth and accelerating sustainable solutions at local level – remain valid. They serve as useful entry points for developing *ESD for 2030* activities.
- 5.11 Second, the five Partner Networks have been instrumental in reaching grassroots actors. The concept deserves continued support,²⁰ but the collaboration among the Partners should be reinforced. To that effect, programme support to each Network should be provided for activities involving different groups of Partners across the network.²¹ Furthermore, that the five Partner Networks should be merged into one inclusive Network of Partners, with five groups of Partners specialized in the five priority action areas included within, but working collaboratively with other groups of Partners. Further efforts will be made to involve representatives of governments, local authorities, the private sector, civil society organizations, the donor and development communities, as well as the SDG communities in the Network of Partners.
- 5.12 Finally, the UNESCO-Japan Prize for ESD, an integral part of the GAP strategy, and which has become one of the most successful prize programmes at UNESCO, deserves further support as an important advocacy tool for ESD and particularly the post-GAP position.
- 5.13 **Activities by Member States:** ESD is an integral element of Target 4.7 of SDG 4 on education. It is included as one of the global education targets for which progress will regularly be monitored. ESD is now well understood as a sine qua non for quality education in relation to the urgency of building a peaceful and sustainable world for the survival and prosperity of current and future generations. To further support ESD as a key enabler of all the other SDGs and especially SDG 4 on education, Member States should make the following efforts.
- 5.14 For priority action area 1 on policy, ESD must be integrated in global, regional and national policies related to education and sustainable development so that these policies can create an enabling environment for pedagogies that support individual empowerment and provide skills for socio-political engagement. Coordinated efforts should be made for ESD to find synergies with other relevant national and international agendas and policies on education and sustainable development.
- 5.15 For priority action area 2 on education and training settings, attention is required to promote the whole-institution approach, emphasizing the importance and necessity for schools or other education institutions, at all levels from early childhood to higher education and lifelong learning in communities, to work together. There have to be strategic policies and measures to reinforce the interaction and cooperation of the formal, non-formal and informal educational settings.
- 5.16 For priority action area 3 on educators, there need to be more opportunities for them to increase their capacities to empower learners. The understanding on how transformative actions occur must be reflected in the capacity-development programmes for formal and non-formal educators so that they are clearly aware of the strengths and weaknesses of the various pedagogical approaches they employ. Educators must be facilitators of learning that guide the learners through the transformation as well as expert transmitters of knowledge.

20 The current membership of the Key Partners Network, however, needs to be renewed, as their terms of reference finished at the end of 2019. A new process to identify partners and the terms of reference for the network will be revised in line with *ESD for 2030*.

21 The partners referred to in this document concern 95 partners identified at the global level within the current phase of the GAP, whose networking activities have been supported by UNESCO.

SOCIAL MEDIA VERIFICATION: ASSESSING SOURCES AND VISUAL CONTENT

Tom Trewinnard and Fergus Bell

(Picture)

MODULE 6

This module is designed to help participants identify and verify the original source of digital information online. It will introduce different strategies for determining the authenticity of sources, photos and videos, especially User Generated Content (UGC) shared via social networks.

By the end of this module, participants should be aware of different types of false and misleading content often shared during breaking news events on platforms such as *Facebook, Twitter, Instagram* and *YouTube*¹. Such content is periodically picked up and carried by otherwise reliable news organisations, serving to discredit them. It is also inadvertently redistributed and amplified on social networks by journalists, who are sometimes targeted by malicious actors with a view to influencing public debate² and leveraging reporters' credibility as trusted sources³.

Participants are asked to test their instincts with real-world scenarios and examples, before putting into practice basic investigative techniques and strategies for verifying content, including:

- ▶ Identifying and crediting original sources in line with the ethical principles guiding journalistic use of User Generated Content⁴
- ▶ Identifying and excluding fake accounts or bots^{5 6}
- ▶ Confirming visual content is correctly attributed to the original source
- ▶ Verifying the recording and upload time of content
- ▶ Geolocating photos and video

Being able to identify and verify original content enables journalists to seek permission to publish User-Generated Content (UGC) in line with ethical and legal requirements.



Outline

The writers Bill Kovach and Tom Rosenstiel, in *The Elements of Journalism*⁷ affirm: “In the end, the discipline of verification is what separates journalism from entertainment,

1 Alejandro, J. (2010). *Journalism In The Age Of Social Media*. Reuters Institute Fellowship. Available at: <https://reutersinstitute.politics.ox.ac.uk/our-research/journalism-age-social-media>.

2 Paulussen, S. & Harder, R. (2014). *Social Media References in Newspapers*. Journalism Practice, 8(5), pp.542-551.

3 Module Seven has a detailed discussion and treatment of this problem

4 See the Online News Association's UGC Ethics guide: <https://ethics.journalists.org/topics/user-generated-content/>.

5 Woolley, S.C. & Howard, P.N. (2017). *Computational Propaganda Worldwide: Executive Summary*. Samuel Woolley and Philip N. Howard, Eds. Working Paper 2017.11. Oxford, UK: Project on Computational Propaganda. comprop.oii.ox.ac.uk. Available at <https://demtech.oii.ox.ac.uk/wp-content/uploads/sites/12/2017/06/Casestudies-ExecutiveSummary.pdf>.

6 Joseph, R. (2018). Guide. *How to verify a Twitter account*. Africa Check. Available at <https://africacheck.org/factsheets/guide-verify-twitter-account/>.

7 Kovach, B., & Rosenstiel, T. (2014). *The elements of journalism: What newspeople should know and the public should expect*. New York: Crown Publishers.

propaganda, fiction, or art.... Journalism alone is focused first on getting what happened down right...”. In this spirit, this Module examines the “discipline of verification” in current times.

Social media has changed journalism practice. Real-time audience engagement has given rise to crowdsourcing content, and even reporting tasks like verification can now be outsourced to the audience.⁸ While journalism remains in essence, a discipline of verification⁹, methods of verifying content and sources require constant updating to reflect the impacts of fast-changing digital technologies, online behaviours, and newsgathering practices. For example, during the Arab Spring, the concept of ‘open verification’ - a process of public, collaborative, real-time verification - began to surface. But this process remains contentious because of the risks associated with misinformation going viral in the course of attempting to verify information step-by-step in a public forum (i.e. a reporter sharing unverified information with a view to crowdsourcing the verification process).¹⁰

Today, eyewitness accounts and visual content are amongst the most important and compelling tools a journalist or news publisher can draw on to tell a high impact story. In a breaking news scenario, speed is a critical factor in verifying information from social media.¹¹

Journalists must be able to navigate vast amounts of information to get to sources, information and images that matter. The rapid growth in the amount of visual content (photos, videos and GIFs) uploaded to social platforms, is driven by three main factors:

- ▶ The proliferation of camera-enabled smart and feature phones around the world¹²
- ▶ Increased access to inexpensive (and in some places, free) mobile data
- ▶ The rise of global social networks and social messaging platforms on which anyone can publish content and build an audience

In many breaking news scenarios, the first accounts, photos and video footage to emerge from an incident – be it a protest, a train crash, a hurricane, or a terrorist attack – are likely to be published by an eyewitness, participant or bystander with a smartphone. Techniques for verifying this content vary depending on newsrooms’ resources, norms and standards, and journalists’ own practices. This module will

8 Carvin, A. (2012). *Distant witness: Social Media's Journalism Revolution*. New York, NY: CUNY Journalism Press.

9 Kovach, B. & Rosenstiel, T. (2014). Op cit

10 Posetti, J. & Silverman, S. (2014). *When Good People Share Bad Things: The basics of social media verification* at Mediashift July 24th, 2014. Available: <http://mediashift.org/2014/07/goodwhen-good-people-share-bad-things-the-basics-of-social-media-verification/>.

11 Brandtzaeg, P., Lüders, M., Spangenberg, J., Rath-Wiggins, L. & Følstad, A. (2015). *Emerging Journalistic Verification Practices Concerning Social Media*. *Journalism Practice*, 10(3), pp.323-342.

12 See Slide 5 of Mary Meeker's Internet Trends Report: <https://www.slideshare.net/DanSchiffman/kleiner-perkins-internet-outlook>.

introduce students to some best practice methods and online tools and resources, but as with the technology, tools are evolving rapidly.¹³

With any verification, some general guidelines, offered by Kovach and Rosenstiel (2014)¹⁴ apply:

- ▶ Edit with scepticism
- ▶ Keep an accuracy checklist
- ▶ Assume nothing – don't be misled by exploitation of signals associated with "truthiness"¹⁵
- ▶ Be cautious with anonymous sources.

By identifying the originator of information or images, and performing a system of checks on both the source and the content they have shared, you should find yourself in a position to verify them as the source, providing the checks give you the required outcomes.¹⁶

These checks replicate the work that a journalist might perform if they were physically present at the scene of a news event interviewing eyewitnesses. A journalist able to conduct an in-person interview would scrutinise the account of the eyewitness, follow up on important details and come to a conclusion about their reliability, based on fact-checking. Instinct can also be a partial guide – along with watching for behaviour clues. The process of confirming a source digitally must allow conclusions to be drawn, even if it is not possible to actually speak to a person directly, or in real-time.¹⁷

Many large newsrooms have teams and expensive technology, or agencies providing services, dedicated to finding this content as quickly as possible¹⁸, while also acquiring the publishing and broadcast rights and verifying the content prior to publication. Most smaller newsrooms and many individual journalists do not have the same resources,¹⁹ and rely on their own evolving, systematic methodology to determine trustworthiness.²⁰

13 Schifferes, S., Newman, N., Thurman, N., Corney, D., Göker, A. & Martin, C. (2014). *Identifying and Verifying News through Social Media*. Digital Journalism, 2(3), pp.406-418.

14 Kovach & Rosenstiel (2014). Op cit.

15 Zimmer, B (2010). "Truthiness", The New York Times. <https://www.nytimes.com/2010/10/2017/magazine/2017FOB-onlanguage-t.html>.

16 Bell, F. (2015). *Verification: Source vs Content* [online] Medium. Available at: <https://medium.com/1st-draft/verfication-source-vs-content-b67d6eed3ado>.

17 Kovach & Rosenstiel (2014). Op cit.

18 Diakopoulos N., De Choudhury M. & Naaman M. (2012). *Finding and assessing social media information sources in the context of journalism* Conference on Human Factors in Computing Systems - Proceedings, , pp. 2451-2460. Available at: <https://dl.acm.org/doi/10.1145/2207676.2208409>.

19 Schifferes, S., Newman, N., Thurman, N., Corney, D., Goker, A.S. & Martin, C. (2014). *Identifying and verifying news through social media: Developing a user-centred tool for professional journalists*. Digital Journalism, 2(3), pp. 406-418. Available at http://openaccess.city.ac.uk/3071/1/IDENTIFYING_AND_VERIFYING_NEWS_THROUGH_SOCIAL_MEDIA.pdf.

20 Brandtzaeg, P. B., Lüders, M., Spangenberg, J., Rath-Wiggins, L., & Følstad, A. (2016). *Emerging journalistic verification practices concerning social media*. Journalism Practice, 10(3), 323-342.

Why is verification of the source and visual content so important? Put simply: it is good journalism. In today's digital world, it is straightforward for ill-intentioned actors to create and share convincing and difficult to detect fakery. There are many cases where professional journalists and newsrooms have damaged their reputation by sharing or republishing misleading information, photos or videos or information from fake persons. At times, they have also misinterpreted satirical content, sharing or publishing it as fact.²¹

The problem is compounded by the volume of visual content available online, all of which can be stripped of context and recycled in future news events, as we see happen daily around the world with hoaxers tricking politicians and professional journalists alike.

There are, however, numerous steps that can be taken to assess the credibility of a given source who has a story to tell, or content to share. Important questions should be asked, some directly, some answered by using the evidence available through investigation. Verification tools can be used to establish where a source has posted from, but it is also possible to manually triangulate a source by analysing their social media history to check for clues that could indicate the feasibility of them being in a particular place at a particular time. Examining the history of their interactions with other users and checking linked content within posts also assists in the manual verification process and can help eliminate information shared by bots.

Sceptical editing is essential, but the vast majority of individuals who get caught up in news events and share their stories are not looking to deceive - they are just sharing their experiences. If misinformation arises, it may not be malicious. Instead it could simply be that the individual cannot remember the events properly or has perhaps chosen to embellish the story. This could also occur if you had the opportunity to conduct a physical, in-person interview, as is frequently borne out in conflicting reports and statements from the scenes of crimes or accidents, where the accounts of traumatised witnesses or victims can vary considerably.

While it might not be possible to ascertain with full certainty the provenance of visual content, there are a number of “red flags” that can be uncovered through a simple verification process that asks:

- ▶ Is the content original, or has it been “scraped” from previous reporting and re-appropriated misleadingly?
- ▶ Has the content been digitally manipulated in some way²²

²¹ Deutsche Welle (2018) *Germany's Bild falls for hoax and unleashes fake news debate* (22/02/2018). Available at <http://www.dw.com/en/germanys-bild-falls-for-hoax-unleashes-debate-on-fake-news/a-42704014>.

²² Student survivors of the mass shooting at a school in Parkland, Florida, USA, who organised a successful national protest for gun control, featured in manipulated images spread on partisan social media channels https://www.buzzfeed.com/janelytyvnenko/here-are-the-hoaxes-and-conspiracies-still-going-around?utm_term=.euy6NPayy#jhe2YvV44.

- ▶ Can we confirm the time and place of the photo/video capture, using available metadata?
- ▶ Can we confirm the time and place of the photo/video capture, using visual clues in the content?

To find red flags efficiently, we also need to understand the different types of common false or misleading visual content:

- ▶ **Wrong time/wrong place:** the most common type of misleading visuals are old visuals that are being re-shared with new claims about what they show. Virality in such cases is often caused by accidental sharing, of content that can be easy to debunk, but not easy to pull back²³
- ▶ **Manipulated content:** content that has been digitally manipulated using photo or video editing software
- ▶ **Staged content:** original content that has been created or shared with the intent of misleading²⁴

In this module, students will be introduced to basic tools and techniques to learn and practice source and content verification (slides, including instructors notes, and additional reading provided) such as:²⁵

Facebook account analysis: Using an online tool from Intel Techniques²⁶ you can find out more about a source by analysing their Facebook account.

Twitter account analysis: Using this guide from Africa Check you can find out more about the source by analysing their social history and thereby identify whether it is a bot tweeting.²⁷

Reverse Image Search: Using either Google Reverse Image Search²⁸, TinEye²⁹ or RevEye³⁰, you can check to see if an image is being recycled to support a new claim or event. Reverse image search lets you see if one or more image databases (with billions of images) contain an earlier version of the image. If reverse image search shows an

23 This video claiming to be evidence of a flood at Bengaluru International Airport in India was in fact a rehashed video from a flood at a Mexican airport.
<https://www.thequint.com/news/webqoof/fake-video-claiming-bengaluru-airport-was-flooded-is-from-mexico>.

24 Artificial intelligence and sophisticated video editing tools make it difficult to discern fake videos, as this footage of Barack Obama shows:
<https://www.youtube.com/watch?v=AmUC4m6w1wo>.

25 Note that news tools continue to evolve and the instructor, together with the learners can discover and test these technologies and techniques.

26 Available at: <https://inteltechniques.com/tools/Facebook.html>.

27 Joseph (2018). Op cit

28 How to do a Google Reverse Image Search: <https://support.google.com/websearch/answer/1325808?hl=en>.

29 Go to <https://www.tineye.com/>.

30 <http://squobble.blogspot.co.uk/2009/12/chromeeye-tineye-extension-for-google.html>.

image existing before a claimed event, this is a major red flag and it is likely the image is recycled from an earlier event. If reverse image search does not return any results, this does not mean the image is original, and you still need to do additional checks.

YouTube Data Viewer: There is no publicly available “reverse video search” - but tools like Amnesty’s YouTube Data Viewer³¹, InVID³² and NewsCheck³³ can detect video thumbnails for YouTube videos, and a reverse image search on those thumbnails can reveal if earlier versions of the video have been uploaded. (Tools also show the exact time of upload.)

EXIF Viewer: EXIF is metadata attached to visual content that includes a wide range of data points created by digital cameras and phone cameras at the point of capture. These can include exact time and date, location metadata, device data, and light setting information. EXIF metadata is thus extremely helpful in the verification process, but a major limitation is that social networks strip the metadata from visual content. This means images shared on Twitter or Facebook will not display EXIF data. If, however, you are able to contact the uploader and acquire the original image file, you can use EXIF data to verify the content. It is also important to note that EXIF data can be modified, so further verification is required.

Participants will get a basic introduction to more advanced techniques, with additional resources for further reading and case studies provided. These techniques include:

- ▶ **Geolocation:** Geolocation is the process of determining where a video or image was captured. This can be straightforward if adequate metadata is available: EXIF data from mobile phones often reveals coordinates, and social content (on Instagram, Facebook and Twitter, for example) is occasionally geotagged (though it is important to note that such metadata is editable and can be misleading). Often, geolocation requires cross-referencing visual characteristics and landmarks from the content with satellite imagery, street-view imagery and visual content available from other sources (such as other visual content posted to Twitter, Instagram, Facebook and YouTube).
- ▶ **Weather corroboration:** Sources such as WolframAlpha³⁴ can reveal historical weather data, allowing us to check if the weather observable in visual content is corroborated by the historical record. (i.e. Is the video showing rain on a day no rain was observed by meteorological sources?)

³¹ How to use Amnesty’s YouTube Data Viewer: <https://citizenevidence.amnestyusa.org/>.

³² InVID video verification tool available at: <http://www.invid-project.eu/tools-and-services/invid-verification-plugin/>.

³³ About NewsCheck: <https://firstdraftnews.org/launching-new-chrome-extension-newscheck/>.

³⁴ WolframAlpha tools available at <https://www.wolframalpha.com/examples/science-and-technology/weather-and-meteorology/>.

- ▶ **Shadow analysis:** One line of investigation into a photo or video is to examine the internal consistency of any visible shadows (i.e. are there shadows where we would expect them to be, and are any visible shadows consistent with relevant light sources?)
- ▶ **Image forensics:** Some tools are able to detect inconsistencies in image metadata that suggest manipulation. The validity of these techniques is highly subject to the context and application, but tools such as Forensically³⁵, Photo Forensics³⁶ and IziTru³⁷ can carry out clone detection and error level analysis that could provide useful insights.



Module Aims

- ▶ To increase awareness of the role of User Generated Content (UGC) shared via social networks in contemporary journalism, along with the risks and pitfalls associated with relying on it
- ▶ To achieve a broad understanding of the importance of securing access and information from the primary source in a story and the process to do so
- ▶ To extend understanding of the need to verify UGC content, and exclude different types of fake and misleading content
- ▶ To increase awareness of the basic methods used to verify images and video, and debunk false visual content



Learning Outcomes

1. A deeper understanding of the role of UGC in contemporary journalism
2. Understanding the need for verification of digital content
3. Awareness of, and a technical understanding of, how to use tools to verify an original source
4. The ability to conduct basic verification steps for photo and video content
5. Awareness of more advanced techniques and metadata that can be used in verification processes
6. Awareness of the need to seek permission to use UGC and other online content and the knowledge of how to do so

³⁵ Wagner, J. (2015). *Forensically, Photo Forensics for the Web*. [Blog] 29a.ch.
Available at: <https://29a.ch/2015/08/16/forensically-photo-forensics-for-the-web>.

³⁶ Fotoforensics tools available at: <http://fotoforensics.com/>.

³⁷ IziTru tools available at: <https://www.izitru.com/>.



Module Format

This module is presented as a 60-minute theoretical lecture and a 120-minute, three-part practical demonstration. However, the practical nature of the subject, lends itself to a longer form interactive workshop with practical exercises to supplement the demonstrations.

Theoretical: Using the notes above, design a lecture dealing with verification as an integral but evolving part of journalism's method in the digital age.

Practical: The 120-minute practical session lends itself to an interactive demonstration and workshop. It can be divided into three distinct parts.

Educators should use the notes above and work through the slides which can be download from the following links. Note there are additional instructor notes attached to the slides:

- i. **Source identification and verification.** Checking social history of sources: https://en.unesco.org/sites/default/files/unesco_fake_news_curriculum_verification_digital_sources_one.pdf
- ii. **Basic image verification.** Common types of false imagery and basic verification steps: https://en.unesco.org/sites/default/files/unesco_fake_news_curriculum_verification_digital_sources_two.pdf
- iii. **More advanced** verification. Approaches for content analysis, including metadata analysis and geolocation: https://en.unesco.org/sites/default/files/unesco_fake_news_curriculum_verification_digital_sources_three.pdf

Linking Plan to Learning Outcomes

A. Theoretical

Module Plan	Number of hours	Learning Outcomes
Lecture: Background & theory on verification and evolution of methods	1 hour	1, 2, 6

B. Practical

Module Plan	Number of hours	Learning Outcomes
i) Source verification - social (Exercise)	30 mins	2, 3
ii) Reverse Image Search (Demonstration & exercise)	15 mins	2, 3, 4
ii) Analysing video (Demonstration)	30 mins	2, 3, 4
iii) Introduction to different types of metadata (Demonstration)	15 mins	2, 5
iii) Geolocation (Demonstration + exercise)	20 mins	2, 4, 5
iii) Weather, shadows and image forensics (Demonstration)	10 mins	2, 4, 5

**Suggested Assignment**

- ▶ Participants should design a source verification workflow using the generic template in slide 8 of the first slide deck. Participants should either use a real role, place they work, or a news organisation that they are familiar with.
- ▶ Select a connected social media account of a popular person and ask participants to use the tools demonstrated to determine whether they are genuine accounts and to identify any related but not authentic accounts.
- ▶ Select and share an image file with the class and ask them to identify certain pieces of information by running it through an online EXIF viewer and a Reverse Image Search tool in order to tell you the original source.

**Materials****Slides**

1. https://en.unesco.org/sites/default/files/unesco_fake_news_curriculum_verification_digital_sources_one.pdf
2. https://en.unesco.org/sites/default/files/unesco_fake_news_curriculum_verification_digital_sources_two.pdf
3. https://en.unesco.org/sites/default/files/unesco_fake_news_curriculum_verification_digital_sources_three.pdf



Reading

Source verification

Ayala Iacucci, A. (2014). *Case Study 3.1: Monitoring and Verifying During the Ukrainian Parliamentary Election*, *Verification Handbook*. European Journalism Centre. Available at: <http://verificationhandbook.com/book/chapter3.1.php>.

Bell, F. (2015). *Verification: Source vs. Content*, First Draft News. Available at: <https://medium.com/1st-draft/verification-source-vs-content-b67d6eed3ado>.

Carvin, A. (2013), *Distant Witness*, CUNY Journalism Press.
Available at: <http://press.journalism.cuny.edu/book/distant-witness-social-media-the-arab-spring-and-a-journalism-revolution/>.

Toler, A. (2017). *Advanced guide on verifying video content*. Available at: <https://www.bellingcat.com/resources/how-tos/2017/06/30/advanced-guide-verifying-video-content/>.

Trewinnard, T. (2016). *Source verification: Beware the bots*, First Draft News. Available at: <https://firstdraftnews.org/articles/source-verification-beware-the-bots/>.

Video

Real or Fake: How to verify what you see on the internet. (2015). France24.
Available at <https://www.youtube.com/watch?v=Q8su4chuU3M&feature=yout>.

Knight, W. (2018). *The Defense Department has produced the first tools for catching deepfakes*, MIT Technology Review. <https://www.technologyreview.com/s/611726/the-defense-department-has-produced-the-first-tools-for-catching-deepfakes/>.

Eyewitness Media

Brown, P. (2015). *A global study of eyewitness media in online newspaper sites*. Eyewitness Media Hub. Available at <https://firstdraftnews.org/articles/user-generated-content/>.

Hermida, A. (2013). *#JOURNALISM*. Digital Journalism, 1(3), pp.295-313.

Koettl, C. (2016, January 27). *Citizen Media Research and Verification: An Analytical Framework for Human Rights Practitioners*. Centre of Governance and Human Rights, University of Cambridge. Available at <https://www.repository.cam.ac.uk/items/a481644f-c1ff-4c71-896f-85e64d156cbo>.

Kuczerawy, A. (2016, December 16). *Pants on fire: content verification tools and other ways to deal with the fake news problem*.

Novak, M. (n.d.). *69 Viral Images From 2016 That Were Totally Fake*. Available at <https://gizmodo.com/69-viral-images-from-2016-that-were-totally-fake-1789400518>.

- Online News Association: UGC Ethics Guide <https://ethics.journalists.org/topics/user-generated-content/>.
- Pierre-Louis, K. (2017). *You're probably terrible at spotting faked photos*. Available at <https://www.popsoci.com/fake-news-manipulated-photo>.
- Rohde, D. (2013). *Pictures That Change History: Why the World Needs Photojournalists*. The Atlantic. Available at <https://www.theatlantic.com/international/archive/2013/12/pictures-that-change-history-why-the-world-needs-photojournalists/282498/>.
- Shapiro, I., Brin, C., Bédard-Brûlé, I. & Mychajlowcz, K. (2013) *Verification as a Strategic Ritual: How journalists retrospectively describe processes for ensuring accuracy*, published in *Journalism Practice*, 7(6).
- Smidt, J. L., Lewis, C. & Schmidt, R. (2017). *Here's A Running List Of Misinformation About Hurricane Irma*. Available at <https://www.buzzfeed.com/janelytynenko/irma-misinfo/>.
- Wardle, C. (2015). *7/7: Comparing the use of eyewitness media 10 years on*. Available at <https://firstdraftnews.org/articles/77-comparing-the-use-of-eyewitness-media-10-years-on/>.
- Wardle, C., Dubberley, S., & Brown, P. (2017). *Amateur Footage: A Global Study of User-Generated Content in TV and Online News Output*. Available at <https://academiccommons.columbia.edu/doi/10.7916/D88S526V>.
- Zdanowicz, C. (2014). *"Miracle on the Hudson" Twitpic changed his life*. Available at <http://www.cnn.com/2014/01/15/tech/hudson-landing-twitpic-krums/index.html>.

Reverse Image Search

First Draft News. *Visual Verification Guide - Photos* -. Available at https://firstdraftnews.org/wp-content/uploads/2017/03/FDN_verificationguide_photos.pdf?x47o84.

First Draft News. *Visual Verification Guide - Video* -. Available at https://firstdraftnews.org/wp-content/uploads/2017/03/FDN_verificationguide_videos.pdf?x47o84.

Suibhne, E. (2015). *Baltimore “looting” tweets show importance of quick and easy image checks*. Available at <https://medium.com/1st-draft/baltimore-looting-tweets-show-importance-of-quick-and-easy-image-checks-a713bbcc275e>.

Seitz, J. (2015). *Manual Reverse Image Search With Google and TinEye*. Available at <https://www.bellingcat.com/resources/how-tos/2015/05/08/manual-reverse-image-search-with-google-and-tineye/>.

YouTube Data Viewer

First Draft News. (n.d.). *Using YouTube Data Viewer to check the upload time of a video* -. Available at <https://www.youtube.com/watch?v=2YS7YWnoYwo>.

Toler, A. (2017). *Advanced Guide on Verifying Video Content*. Available at <https://www.bellingcat.com/resources/how-tos/2017/06/30/advanced-guide-verifying-video-content/>.

Metadata Analysis

Honan, M. (2012). *How Trusting in Vice Led to John McAfee’s Downfall*. Available at <https://www.wired.com/2012/12/how-vice-got-john-mcafee-caught/>.

Storyful. (2014). *Verifying images: why seeing is not always believing*.

Wen, T. (2017). *The hidden signs that can reveal a fake photo*. Available at <http://www.bbc.com/future/story/20170629-the-hidden-signs-that-can-reveal-if-a-photo-is-fake>.

Content Analysis

Ess, H. van. (2017). *Inside the trenches of an information war.*

Medium. Available at <https://medium.com/@henkvaness/how-to-date-a-mysterious-missile-launcher-78352ca8c3c3>.

First Draft News. (2016). *Watch Eliot Higgins demonstrate advanced verification techniques at #FDLive*. Available at <https://www.youtube.com/watch?v=D7zvsXv2ECI&list=PLon8am2uBRCDS8-4WP7EiyGQQgYW7sEoV&index=2>.

Higgins, E. (2015, July 24). *Searching the Earth: Essential geolocation tools for verification*. Available at <https://medium.com/1st-draft/searching-the-earth-essential-geolocation-tools-for-verification-89d96obb8fba>.

Online resources

First Draft Interactive: Geolocation Challenge. Available at https://ftp.firstdraftnews.org/articulate/glch318/story_html5.html.

First Draft Interactive: Observation Challenge. Available at https://ftp.firstdraftnews.org/articulate/2020/en/Oc20/story_html5.html.

First Draft Online Verification Course. Available at <https://firstdraftnews.org/training/verification/>.

COMBATting ONLINE ABUSE: WHEN JOURNALISTS AND THEIR SOURCES ARE TARGETED

Julie Posetti

(Picture)

MODULE 7



Synopsis

The problem of disinformation and misinformation¹ undermining credible journalism and reliable information has escalated dramatically in the social media era. Consequences include the deliberate targeting of journalists and other online publishers, along with their sources, who are seeking to verify or share information and commentary. The associated risks can further undermine trust in journalism, along with the safety of journalists and their sources.

In some instances, journalists have been targeted in acts of ‘astroturfing’² and ‘trolling’³ - deliberate attempts to “mislead, misinform, befuddle, or endanger journalists”⁴ with the sharing of information designed to distract and misdirect them, or their potential sources. Alternatively, journalists might be targeted to trick them into sharing inaccurate information which feeds a false interpretation of the facts or, when it is revealed as fake, diminishes the credibility of the journalist (and the news organisation with which they are affiliated). In other cases, they face digital threats designed to expose their sources, breach their privacy to expose them to risk, or access their unpublished data.

There is also the phenomenon of governments mobilising ‘digital hate squads’ to chill critical commentary and quash freedom of expression.⁵ Then, there is the serious problem of online harassment and violence (sometimes problematically labelled as ‘trolling’⁶) disproportionately experienced by women and frequently misogynistic in nature. This can see journalists, their sources, and commentators subjected to torrents of online abuse, false claims about their conduct, misrepresentation of their identities, or threats of harm designed to humiliate them and undermine their confidence, discredit them, divert their attention and, ultimately, to chill their reporting.⁷ Meanwhile in many places, physical world abuse designed to suppress critical reporting continues, with the added danger of now being fuelled by online incitement and intimidation.

1 For definitions, see: Wardle, C. & Derakhshan, H. (2017). *Information Disorder: Toward an Interdisciplinary Framework for Research and Policymaking* (Council of Europe). <https://rm.coe.int/information-disorder-toward-an-interdisciplinary-framework-for-research/168076277c>.

2 ‘Astroturfing’ is a term derived from a brand of fake grass used to carpet outdoor surfaces to create the impression it is natural grass cover. In the context of disinformation, it involves spreading fake information, targeting audiences and journalists with an intention to redirect or mislead them, particularly in the form of ‘evidence’ of faux popular support for a person, idea or policy. See also Technopedia definition: <https://www.techopedia.com/definition/13920/astroturfing>.

3 Coco, G. (2012). *Why Does Nobody Know What Trolling Means? A quick reference guide for the media* at Vice.com. https://www.vice.com/en_au/article/ppqk78/what-trolling-means-definition-UK-newspapers.

4 Posetti, J. (2013). *The ‘Twitterisation’ of investigative journalism* in S. Tanner & N. Richardson (Eds.), *Journalism Research and Investigation in a Digital World* (pp. 88-100): Oxford University Press, Melbourne. https://ro.uow.edu.au/articles/chapter/The_twitterisation_of_investigative_journalism/27777093.

5 Riley M, Etter, L and Pradhan, B (2018) *A Global Guide To State-Sponsored Trolling*, Bloomberg: <https://www.bloomberg.com/features/2018-government-sponsored-cyber-militia-cookbook/>.

6 Note: ‘Trolling’ in its internet-related application refers to acts that range from gentle teasing, tricking and goading to deliberate deception. However, it is increasingly deployed as a term to cover all acts of online abuse. This is potentially problematic as it conflates a wide range of activities and potentially underplays the seriousness of online harassment.

7 See for example: <https://www.independent.co.uk/news/world/americas/twitter-maggie-haberman-new-york-times-quits-social-media-jack-dorsey-a8459121.html>

Journalists can be direct victims of disinformation campaigns, but they are also pushing back. In addition to strengthening digital defences, many are proactively exposing these attacks and uncovering the attackers. Engaging in Media and Information Literacy initiatives along with NGOs in this space, news media are also playing a role in educating the public about why journalism is worth cherishing and protecting.



Outline

Teasing out the issues

i) Recognising and Responding to ‘Trolling’ and ‘Astroturfing’⁸

This phenomenon includes fabrication of characters and events designed to trick journalists and audiences, along with organised social media campaigns aimed at mimicking organic public reaction. It can be difficult to differentiate breaking news and legitimate witness accounts from content that has been faked or peppered with inaccuracies to deliberately mislead or undermine the credibility of journalists and other online commentators, along with their work, by duping them into sharing false information.

Examples of this kind of behaviour include:

- ▶ The fabrication of disaster victims and terrorist attack casualties (see Manchester bombings example⁹) to fool people into sharing content that potentially damages the reputation and/or credibility of individuals, including journalists, who might be tagged in the distribution process.
- ▶ The publication of content parading as newsworthy produced by fictitious characters such as the ‘Gay girl in Damascus’¹⁰. In 2011, the world’s media clamoured to report the arrest of the blogger who was purportedly a Syrian lesbian - the author turned out to be a U.S. student based outside the country. Journalist Jess Hill was assigned to the story for the Australian Broadcasting Corporation’s PM programme. She says traditional verification values and methods prevented her programme from amplifying a falsehood. “We didn’t report her arrest, for one simple reason – we couldn’t find anyone who had actually met her in person. No relatives, no personal friends. We spent two days looking for people, asking our Syrian contacts to refer us to people who may have had contact with her, but each lead became a dead end. The fact that we couldn’t find anyone who had actually met her set off major alarm bells, so we didn’t report it... News agencies who rushed to

⁸ For an explanation of ‘astroturfing’ useful for teaching purposes, the following link is of value: <https://youtu.be/Fmh4RdlwswE>

⁹ Manchester bombing example, <https://www.theguardian.com/technology/2017/may/26/the-story-behind-the-fake-manchester-attack-victims>.

¹⁰ Young, K. (2017). *How to Hoax Yourself: The Case of the Gay Girl in Damascus*, November 9th, 2017, in The New Yorker. <https://www.newyorker.com/books/page-turner/how-to-hoax-yourself-gay-girl-in-damascus>.

report that story didn't do the basic job of going back to the source. They reported news based on an entry on a blog."¹¹

Other motivations include the desire to redirect or distract journalists from an investigation by prompting fruitless lines of inquiry that stymie reporting efforts and, ultimately, have a chilling effect on truth-seeking.

Examples of this style of misdirection include:

- ▶ The attempted reframing of claims about the size of the crowd at Donald Trump's inauguration in January 2017 as 'alternative facts'¹²
- ▶ Contemporary wartime propaganda, e.g. the Taliban tweeting at journalists in Afghanistan with false and misleading details of battles.¹³
- ▶ Datasets handed to journalists that provide some verifiable public interest-value information but have been corrupted by disinformation in the mix.

More recently, computational propaganda¹⁴ has increased the risks for journalists dealing with 'astroturfing' and 'trolling'. This involves the use of bots to disseminate well-targeted false information and propaganda messages on a scale designed to look like an organic movement.¹⁵ Concurrently, AI technology is being leveraged to create 'deepfake'¹⁶ videos and other forms of content designed to discredit the targets, including journalists, and especially female reporters.

Examples of these practices include:

- ▶ Independent news site Rappler.com and its largely female staff were targeted in a campaign of prolific online abuse. "In the Philippines, paid trolls, fallacious reasoning, leaps in logic, poisoning the well – these are only some of the propaganda techniques that have helped shift public opinion on key issues."¹⁷ (see expanded discussion below)

¹¹ Posetti, J. (2013). op cit

¹² NBC News (2017) Video: <https://www.nbcnews.com/meet-the-press/video/conway-press-secretary-gave-alternative-facts-860142147643>.

¹³ Cunningham, E (2011). *In shift, Taliban embrace new media*, GlobalPost. <https://www.pri.org/stories/2011-05-21/shift-taliban-embrace-new-media>.

¹⁴ Woolley, S. & Howard, P. (2017). *Computational Propaganda Worldwide: Executive Summary*, Working Paper No. 2017.11 (Oxford University). <https://demotech.oii.ox.ac.uk/wp-content/uploads/sites/12/2017/06/Casestudies-ExecutiveSummary.pdf>.

¹⁵ Note: Shallow reports about bot campaigns during the 2017 UK general election highlight the difficulty of reporting on these issues. C.f. Dias, N. (2017). *Reporting on a new age of digital astroturfing*, First Draft News. <https://firstdraftnews.org/articles/digital-astroturfing/>.

¹⁶ The term deepfake is a portmanteau of 'deep learning' and 'fake'. It involves AI technology in the creation of fraudulent content, sometimes of a pornographic nature, that is virtually undetectable. It is used in cyberattacks to discredit people, including journalists. See: Cuthbertson, A (2018) *What is 'deepfake' porn? AI brings face-swapping to disturbing new level* in Newsweek <http://www.newsweek.com/what-deepfake-porn-ai-brings-face-swapping-disturbing-new-level-801328>.

¹⁷ Ressa, M. (2016). *Propaganda War: Weaponising the Internet*, Rappler. <https://www.rappler.com/nation/148007-propaganda-war-weaponizing-internet>.

- ▶ A wealthy family accused of capturing key state enterprises and politicians in South Africa hired UK Public Relations firm Bell Pottinger to devise an elaborate propaganda campaign. It spread its messages via a disinformation empire involving websites, media and a paid Twitter army which targeted journalists, business people and politicians with abusive, hostile messages and photoshopped images, designed to humiliate and counter their investigations into state capture.¹⁸ Prominent editor Ferial Haffajee was targeted in a campaign of online harassment during this period, which saw her image manipulated to create false impressions of her character, alongside deployment of the hashtag #presstitute¹⁹
- ▶ The case of journalist Rana Ayyub elicited a call by five United Nations special rapporteurs for the Indian government to provide protection, following the mass circulation of false information aimed to counter her critical reporting. The independent journalist had been on the receiving end of a combination of disinformation about her on social media, including ‘deepfake’ videos that falsely suggested she had made pornographic films, as well as direct rape and death threats²⁰
- ▶ The case of Finnish journalist, Jessikka Aro, discussed under ‘Digital Safety Threats and Defensive Strategies’ in section ii) of this module.

Other modules in this handbook deal specifically with technical verification techniques, but it is important to enable participants to identify the malicious motivation of some online operators in the creation, distribution and targeting of journalists with disinformation and misinformation as part of a pattern of abuse.

Critical questions to add to technical methods of information verification:

1. Could there be malicious intent behind this share or tag?
2. What does the person posting the content stand to gain by sharing?
3. What could be the consequences for me/my professional credibility/a news media institution or employer if I share it?
4. Have I worked hard enough to ascertain this individual’s identity/affiliations/reliability/motivations (e.g. are they seeking to seed disinformation or profit from the sale of content acquired illegally without public interest justification)?

¹⁸ Extensive dossier on the Gupta’s ‘fake news empire’ available at <https://www.timeslive.co.za/news/south-africa/2017-09-04-the-guptas-bell-pottinger-and-the-fake-news-propaganda-machine/>.

¹⁹ Haffajee, F. (2017). *Ferial Haffajee: The Gupta fake news factory and me*. HuffPost South Africa. [online] Available at: https://www.huffingtonpost.co.uk/entry/ferial-haffajee-the-gupta-fake-news-factory-and-me_uk_5c7eaa44e4b078abc6c24e7f.

²⁰ UN experts call on India to protect journalist Rana Ayyub from online hate campaign <http://www.ohchr.org/EN/NewsEvents/Pages/DisplayNews.aspx?NewsID=23126&LangID=E>; See also Ayyub, R. (2018). *In India, journalists face slut-shaming and rape threats*. <https://www.nytimes.com/2018/05/22/opinion/india-journalists-slut-shaming-rape.html>.

5. Is this a human or bot?
6. If you receive a 'data dump' from a purported whistleblower, should you independently verify the contents before publishing the dataset in full? Is it possible that it is peppered with disinformation and misinformation designed to deliberately mislead or discredit?

ii) Digital Safety Threats and Defensive Strategies

Journalists, human rights defenders and bloggers/social media activists are increasingly vulnerable to cyber-attacks, and their data or sources may be compromised by malicious actors including through phishing, malware attacks, and identity spoofing.²¹

An example of this practice:

Award-winning investigative journalist Jessikka Aro, who works for Finland's public broadcaster YLE, has been the target of organised 'troll' campaigns since 2014. She has experienced digital safety threats including spoofing and doxing²², with trolls disclosing her personal contact information and spreading disinformation about her, rendering her messaging apps and inboxes full of angry messages. *"I received a phone call in which someone fired a gun. Later, someone texted me, claiming to be my dead father and told me he was 'observing' me,"* she says.²³ Aro has expressed appreciation for editors who protect journalists from threats and urged journalists to investigate and expose propaganda.

It is therefore important for journalistic actors to be alert to the following threats:

12 key digital security threats²⁴

- ▶ Targeted surveillance and mass surveillance
- ▶ Software and hardware exploits without the knowledge of the target
- ▶ Phishing attacks²⁵
- ▶ Fake domain attacks

21 From Technopedia: Spoofing is a fraudulent or malicious practice in which communication is sent from an unknown source disguised as a source known to the receiver. Email spoofing is the most common form of this practice. A spoofed email may also contain additional threats like Trojans or other viruses. These programmes can cause significant computer damage by triggering unexpected activities, remote access, deletion of files and more <https://www.techopedia.com/definition/5398/spoofing>.

22 From Technopedia: Doxing is the process of retrieving, hacking and publishing other people's information such as names, addresses, phone numbers and credit card details. Doxing may be targeted toward a specific person or an organisation. There are many reasons for doxing, but one of the most popular is coercion. Doxing is a slang term that is derived from the word ".doc" because documents are often retrieved and shared. Hackers have developed different ways to dox, but one of the most common methods is by obtaining the victim's email and then uncovering the password to open their account to obtain more personal information. <https://www.techopedia.com/definition/29025/doxing>.

23 Aro, J. 2016. *The cyberspace war: propaganda and trolling as warfare tools*. European View. Sage Journals, June 2016, Volume 15, Issue 1. <http://journals.sagepub.com/doi/full/10.1007/s12290-016-0395-5>.

24 Posetti, J. (2015). New Study: *Combating the rising threats to journalists' digital safety* (WAN-IFRA). <https://blog.wan-ifra.org/2015/03/27/new-study-combating-the-rising-threats-to-journalists-digital-safety>.

25 King, G (2014) *Spear phishing attacks underscore necessity of digital vigilance*, CPJ. <https://cpj.org/blog/2014/11/spear-phishing-attacks-underscore-necessity-of-dig.php>.

- ▶ Man-in-the-Middle (MitM) attacks²⁶
- ▶ Denial of Service (DoS) attacks and Distributed Denial of Service (DDoS – Distributed Denial of Service)²⁷
- ▶ Website defacement
- ▶ Compromised user accounts
- ▶ Intimidation, harassment and forced exposure of online networks
- ▶ Disinformation and smear campaigns
- ▶ Confiscation of journalistic work product, and
- ▶ Data storage and mining

For defensive strategies see: *Building Digital Safety for Journalism*.²⁸

For the implications for confidential sources and whistleblowers interacting with journalists and other media producers see: *Protecting Journalism Sources in the Digital Age*.²⁹

Recognising and managing online harassment and violence

“I’ve been called a dirty whore, a bloody Gypsy, Jewish, a Muslim slut, a Greek parasite, a disgusting migrant, a stupid psycho, an ugly liar, a biased hater. They keep telling me to go home, to kill myself or they will shoot me, cut my tongue off, break my fingers one by one. They keep threatening me with gang rapes and sexual torture.”³⁰ These are the words of celebrated Swedish journalist Alexandra Pascalidou, who testified in 2016 before a European Commission session in Brussels about her experiences online.

The global proliferation of this kind of online abuse targeting women journalists and commentators has led to the UN (including UNESCO³¹) and other agencies recognising the problem, and calling for actions and solutions.

The Organisation for Security and Co-operation in Europe (OSCE) has sponsored research that demonstrates the international impact of online abuse of female

26 Technopedia definition of Man in the Middle Attack (MITM): “A form of eavesdropping where communication between two users is monitored and modified by an unauthorised party. Generally, the attacker actively eavesdrops by intercepting a public key message exchange and retransmits the message while replacing the requested key with his own”. <https://www.techopedia.com/definition/4018/man-in-the-middle-attack-mitm>.

27 See definitions at Technopedia. <https://www.techopedia.com/definition/24841/denial-of-service-attack-dos> b. <https://www.techopedia.com/definition/10261/distributed-denial-of-service-ddos>.

28 Henriksen, J. et al. (2015). *Building Digital Safety for Journalism* (UNESCO) Paris. <http://unesdoc.unesco.org/images/0023/002323/232358e.pdf>.

29 Posetti, J. (2017). *Protecting Journalism Sources in the Digital Age* (UNESCO). Paris. <http://unesdoc.unesco.org/images/0024/002480/248054E.pdf>.

30 Posetti, J. (2016). *Swedish journalist Alexandra Pascalidou Describes Online Threats of Sexual Torture and Graphic Abuse in The Sydney Morning Herald*, 24/11/2016. <http://www.smh.com.au/lifestyle/news-and-views/swedish-broadcaster-alexandra-pascalidou-describes-online-threats-of-sexual-torture-and-graphic-abuse-20161124-gswuwv.html>.

31 Posetti, J. (2017). *Fighting Back Against Proliferating Online Harassment: Maria Ressa in L. Kilman (Ed) op cit* See also: Resolution 39 of UNESCO’s 39th General Conference which notes “the specific threats faced by women journalists including sexual harassment and violence, both online and offline.” <http://unesdoc.unesco.org/images/0026/002608/260889e.pdf>.

journalists who are disproportionately targeted for ‘hate trolling’.³²

That research followed a study by British think tank, Demos, which examined hundreds of thousands of tweets and found journalism was the only category where women received more abuse than men, “with female journalists and TV news presenters receiving roughly three times as much abuse³³ as their male counterparts”. The keywords for the abusers were «slut», «rape» and «whore».

A hallmark of this online abuse of female journalistic actors is the use of disinformation tactics – lies are spread about their character or their work as a means of undermining their credibility, humiliating them, and seeking to chill their public commentary and reporting.

The addition of threats of violence, including rape and murder, and the ‘pile on’ effect (organic, organised, or robotic mass attacks against a person online) worsen the impacts.

The intimate nature of these attacks, often received on personal devices first thing in the morning and last thing at night, further sharpens the impact. “There are days when I wake up to verbal violence and fall asleep with sexist and racist rage echoing in my ears. It’s like a low-intense, constant warfare”, Pascalidou says.

In the Philippines, *Rappler* CEO and Executive Editor, Maria Ressa³⁴, is a case study in combatting prolific online harassment in the context of a massive disinformation campaign with links to the State. She is a former *CNN* war correspondent but she says none of her experiences in the field prepared her for the massive and destructive campaign of gendered online harassment that has been directed at her since 2016. “I’ve been called ugly, a dog, a snake, threatened with rape and murder,” she says. Ressa has lost count of the number of times she has received death threats. In addition, she has been the subject of hashtag campaigns like #ArrestMariaRessa and #BringHerToTheSenate, designed to whip-up online mobs into attack mode, discredit both Ressa and *Rappler*, and chill their reporting. “It began a spiral of silence. Anyone who was critical or asked questions about extrajudicial killings was attacked, brutally attacked. The women got it worst. And we’ve realised that the system is set up to silence dissent - designed to make journalists docile. We’re not supposed to be asking hard questions, and we’re certainly not supposed to be critical,” Ressa says.³⁵

Maria Ressa’s fightback strategy includes:

► Recognising the seriousness of the problem

32 OSCE (2016). *Countering Online Abuse of Female Journalists*. <http://www.osce.org/fom/220411?download=true>.

33 Bartlett, J. et al. (2014) *Misogyny on Twitter*, Demos. https://www.demos.co.uk/files/MISOGYNY_ON_TWITTER.pdf.

34 Maria Ressa is chair of the jury of the UNESCO-Guillermo Cano World Press Freedom Prize <https://web.archive.unesco.org/20210405005819/https://en.unesco.org/prizes/guillermo-cano/jury>.

35 Posetti, J. (2017). *Fighting Back Against Prolific Online Harassment: Maria Ressa* in Kilman, L. (Ed) *An Attack on One is an Attack on All* (UNESCO). <http://unesdoc.unesco.org/images/0025/002593/259399e.pdf>.

- ▶ Recognising the psychological impacts and facilitating psychological support for affected staff
- ▶ Using investigative journalism as a weapon in the fightback³⁶
- ▶ Asking loyal audiences to help repel and contain attacks
- ▶ Tightening security on and offline in response to harassment
- ▶ Publicly calling on platforms (e.g. Facebook and Twitter) to do more to curtail and adequately manage online harassment

While dealing with the rising threat of online harassment, it is also important to acknowledge ongoing offline harassment of women journalists in the context of disinformation campaigns. For example, Australian investigative journalist Wendy Carlisle was abused, heckled and jostled during a climate change denialists' rally in Australia in 2011 while making a documentary for ABC Radio. The abuse led her to leave the event to ensure her safety.³⁷



Module Aims

This module will: inform participants about the risks of online abuse in the context of 'information disorder'; help participants to recognise threats; and provide skills development and tools to assist in combatting online abuse. The aims are:

- ▶ To increase participants' awareness of the problem of malicious actors targeting journalists, their sources, and other online communicators in disinformation/misinformation campaigns;
- ▶ To enable participants to better recognise 'astroturfing', 'trolling', digital safety threats, and online abuse,
- ▶ To equip participants to be better prepared to combat 'astroturfing' and 'trolling', digital safety threats, and online abuse in a gender sensitive manner.



Learning Outcomes

By the end of this module, participants will:

1. Have a deeper understanding of the impacts of online abuse on journalistic actors, journalism, information sharing, and freedom of expression;

³⁶ This was also a tactic deployed by Ferial Haffajee in the 'Gupta leaks' case study referenced earlier. She used investigative journalism techniques and digital security 'detectives' to unmask some of the trolls who had been targeting her in an effort to discredit her reporting of the scandal. See: <https://www.news24.com/SouthAfrica/News/fake-news-peddlers-can-be-traced-hawks-20170123>.

³⁷ Carlisle, W. (2011). *The Lord Monckton Roadshow*, Background Briefing, ABC Radio National. <http://www.abc.net.au/radionational/programs/backgroundbriefing/the-lord-monckton-roadshow/2923400>.

2. Be more aware of the problem of malicious actors targeting journalists and other online communicators in disinformation/misinformation campaigns;
3. Understand the particular safety threats confronting women doing acts of journalism online;
4. Be able to more easily recognise malicious actors online, along with incidents of ‘astroturfing’, ‘trolling’, digital safety threats, and online abuse;
5. Be better equipped to combat ‘astroturfing’, ‘trolling’, digital safety threats, and online abuse in a gender sensitive manner.



Module Format

This module is designed to be delivered face-to-face or online. It is intended for execution in two parts: Theoretical and practical.

Linking Plan to Learning Outcomes

A. Theoretical

Module Plan	Number of hours	Learning Outcomes
An interactive lecture and Q&A (90 minutes), which could be delivered traditionally, or via a webinar platform, designed to encourage remote participation. Lecture content can be drawn from the theory and examples supplied above. However, course convenors are encouraged to also include culturally/locally relevant case studies in the delivery of this module.	60 - 90 mins	1, 2, 3, 4, 5

B. Practical

Module Plan	Number of hours	Learning Outcomes
A workshop/tutorial (90 minutes) which could be facilitated in a traditional classroom setting, or via an eLearning platform like Moodle, Facebook groups or other services that enable remote online participation. The workshop/tutorial exercise could adopt the following format: • Divide tutorials into working groups of 3-5 participants each • Each working group is to be provided with an example of malicious content (Search blogs and social media channels for content created to target Maria Ressa, Jessikka Aro, and Alexandra Pascalidou, for example, whose cases are discussed in this module) connected to a dis/misinformation/trolling/astroturfing/online abuse campaign. • Each working group must: collaboratively assess the material (research the individual/group behind the material); identify risks and threats (referring to relevant research about impacts contained in recommended readings); propose a plan of action for responding to the material (this could include replying strategically, reporting the user to the platform or police if appropriate, assigning a story on the issue); write a 250 word summary of their plan of action (using Google Docs or a similar collaborative editing tool) and submit to their lecturer/tutor for review	90 - 120 mins	1, 2, 3, 4, 5

Alternative structure

For deeper treatment of the issues, this module could be expanded to run as three separate lessons (each delivered in two parts, as described above):

- ▶ Recognising and responding to ‘trolling’ and ‘astroturfing’
- ▶ Digital threat-modelling³⁸ and defensive strategies
- ▶ Recognising and managing gendered online harassment and violence

³⁸ Stray, J. (2014). *Security for journalists, Part Two: Threat Modelling*. <https://source.opennews.org/articles/security-journalists-part-two-threat-modeling/>.

Online resources

VIDEO: *How to Tackle Trolls and Manage Online Harassment* – a panel discussion at the International Journalism Festival, Perugia, Italy (April 2017) with Julie Posetti (Fairfax Media), Hannah Storm (International News Safety Institute), Alexandra Pascalidou (Swedish journalist), Mary Hamilton (*The Guardian*), Blathnaid Healy (CNNi). Available at: <http://media.journalismfestival.com/programme/2017/managing-gendered-online-harrassment>